AI PRODUCT CASE STUDY · INSURANCE · US + INDIA

The claim is the product

How I would build a GenAI claims assistant for personal lines: report on WhatsApp or by text, get paid in minutes when the claim is simple, and get a real person when it isn't.

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A self-initiated case study, not a product I shipped. Where I draw on my own work, it is from my time as a PM in motor insurance at ICICI Lombard. Small teal numbers point to the sources on the last page; my own estimates are labelled as estimates.

The problem	People buy insurance for one moment, the claim, and that is where we make them do the most work. In India, "insurer not disposed of the claim" was the top complaint reason in FY25 [17]. In the US, the average repairable car took 19.3 days to get back [5].
The bet	Most personal-lines claims are simple. An assistant that understands photos, voice notes and documents can file them in one conversation, and the clean, small ones can be paid straight through.
MVP	India: private-car own-damage claims under ₹50,000, on WhatsApp, in Hindi, Telugu and English. US: drivable auto damage and glass, by text. Three months in shadow mode before any automated payout.
My one rule	AI can say yes. Only a person can say no.

01 Why this problem

Since 2023 I have been building AI for motor insurance: AI-led vehicle inspection at policy issuance, ML fraud models inside claims, and a claim admissibility engine with 20+ automated checks. The thing I kept noticing is that the claim is where a customer decides whether insurance was worth paying for, and it is exactly where we hand them the most forms.

In the US, the pain is time and hand-offs. J.D. Power's 2025 auto claims study put the average repair cycle at 19.3 days, down from 22.3 a year earlier [5]. For home claims, filing to finished repairs averages 32.4 days [6]. The average collision claim was $5,489 in 2024 [3]. That is real money for a family, and they spend those weeks chasing status and repeating their story.

In India, the pain is uncertainty and paperwork. Motor was 24.8% of all insurance complaints in FY25 [14]. "Insurer not disposed of the claim" was the biggest single complaint reason at 15.07%, and "claim repudiated without giving reasons" more than doubled, from 3.15% in FY23 to 6.54% in FY25 [17]. On the health side, insurers settled 3.26 crore claims in FY25, and about 1.2 crore of them were reimbursement claims [15]. Each one of those starts with someone photographing a stack of hospital bills.

The Indian regulator has turned speed into a compliance problem, not just a CX one. IRDAI's 2024 master circular says a surveyor must be allocated within 24 hours of a claim, the survey report is due in 15 days, the insurer must decide within 7 days of it, and the claimant gets ₹500 a day if the surveyor is late [13]. Cashless health requests have to be decided within one hour, and final discharge approval within three [16]. And no claim can be rejected for want of documents [13].

> **That last rule shapes the product.** The assistant has to collect everything in one go, and it can never close a claim just because the customer went quiet. Chasing documents is our job, not theirs.

02 Who I am building for

Ravi, 29, Hyderabad. Scraped his hatchback against a pillar in his office basement. Thinks in Telugu, types in English, lives on WhatsApp. Has never heard the words “intimation” or “surveyor”. He wants three answers: am I covered, how much will I get, and can I go to my own garage.

Lakshmi, 52, Vijayawada. Paid for her husband’s surgery at a hospital outside her insurer’s network. She has 40 pages of bills, a discharge summary and a pharmacy file. Her son usually handles the “online stuff”, but he lives in Pune.

Megan, 34, Ohio. Rear-ended at a light, car still drivable, two kids in the back seat. She wants to file from the parking lot, know whether she gets a rental, and not tell the story to three different people.

The adjuster or surveyor. Has a full queue and spends a large part of the day chasing documents and re-typing what the customer already said. Does not want a robot making decisions. Does want the paperwork done, so they can spend their time on claims that need judgement.

One more user that is easy to forget in India: the **agent or garage** that files on the customer’s behalf. Many first reports will come through them, so the assistant needs an “I’m filing for someone else” mode from day one.

03 Why AI here, and where I would keep it out

Claims intake is an unstructured-data problem. A voice note in Hinglish, six blurry photos, a PDF estimate, a scanned registration certificate. Until recently, turning that into a structured claim needed a person. Multimodal models change that, and that is the whole reason this product makes sense now.

Use AI for	Keep AI out of
Understanding what happened, from text, voice and photos, in the customer’s language	**Coverage decisions.** Policy terms are rules. They live in a deterministic engine that can be audited line by line.
Reading documents and pulling out fields, with a confidence score on each	**Any denial or reduction.** A licensed person decides, and the customer gets the reason.
Photo-based damage estimates for small, visible damage	**Injury and third-party claims.** In India these go to motor accident tribunals; in the US they are litigation-prone.
Spotting fraud patterns across thousands of claims	**Total loss and theft.** Low volume, high value, high fraud risk.
Summarising a file for the adjuster, with every fact linked to its source	**Moving money.** Payee checks and bank verification stay deterministic.

From my work on fraud models, the lesson I would carry over is simple: a model score is very good at deciding which queue a claim goes to. It is a poor reason to tell a customer no.

04 How big is this

United States

US insurers wrote $182.4B in personal auto liability, $158.5B in personal auto physical damage and $173.2B in homeowners premium in 2024, about $514B together [1]. Across P&C, “adjusting and other expenses” ran at 5.5% of earned premium in 2024 [2]. If that ratio roughly holds for personal lines, carriers spend **around $28B a year just handling these claims (my estimate).** For scale: 5.3% of insured homes had a claim in 2023 [4].

The part I can serve is the simple claims. If 30% of personal-lines claims are simple enough for an assisted or straight-through path, and the assistant takes 40% of the handling effort out of those, that is **about $3.4B a year of addressable cost (estimate; both percentages are my assumptions).** The first job in a pilot is to replace them with a real carrier's claim mix.

Is the upside real? Lemonade says 55% of its claims are automated and 95% arrive through a digital first notice of loss with AI built in [7]. That is a digital-native carrier with simple products, so I treat it as a ceiling, not a forecast for a traditional insurer.

India

Motor premium was ₹99,093 crore and health (including personal accident) ₹1,27,417 crore in FY25 [15]. Health alone produced 3.26 crore settled claims worth ₹94,248 crore, about ₹28,910 per claim. 58% were cashless and 69% ran through TPAs [15].

The economics flip. Tickets are small, so cost to serve each claim has to be tiny. That pushes me to WhatsApp, cheaper models, and as much straight-through processing as the regulator allows. The upside is reach: WhatsApp has about 500 million users in India [18].

> **A regulatory gift.** Motor losses below ₹50,000 do not need a licensed surveyor [13], and insurers already use app-based AI assessment for them [14]. The regulator has effectively drawn my straight-through lane for me.

05 The product: Claim Companion

One assistant, four steps. The customer only ever sees a conversation. Behind it, the work is split between models that understand and rules that decide.

- **Report.** Start wherever you are: WhatsApp in India, a text link in the US, a phone call, or an agent or garage files for you. Describe what happened in your own words, typed or spoken.
- **Understand.** The assistant finds the policy from a registration number or phone number, asks only for what is missing, and guides a live-camera photo capture. It reads documents as they arrive and reads the key facts back: "Front bumper and left headlamp, at your office in HITEC City, yesterday around 6 pm. Right?"
- **Decide.** Extraction and damage models feed a deterministic coverage engine. Fraud signals choose the lane, never the outcome.
- **Act.** Clean, small claims are paid in minutes. Everything else lands on an adjuster's desk with a one-page summary, each fact linked to its source. Status updates go out without the customer asking.

Three design choices I would defend in a review

1. Ask once, completely. One checklist up front, built for this specific claim, instead of a drip of "please also send". IRDAI bars rejection for missing documents [13], and every extra request is one more chance for the customer to give up. When I worked on DIY inspection, adoption went up 30% after we fixed the friction points in the self-serve flow. In self-serve, the experience around the model matters as much as the model.

2. Photos through a live camera, not the gallery. Forwarded images lose the metadata I want for forensics and are easy to edit. On WhatsApp, the photo step opens a light in-browser camera link and comes straight back to the chat.

3. Every "no" is explained. If an adjuster cuts or denies a claim, the assistant sends the reason in plain words, in the customer's language, with the policy clause. "Repudiated without reasons" is a fast-growing complaint in India [17]. This is the cheapest fix in the whole product.

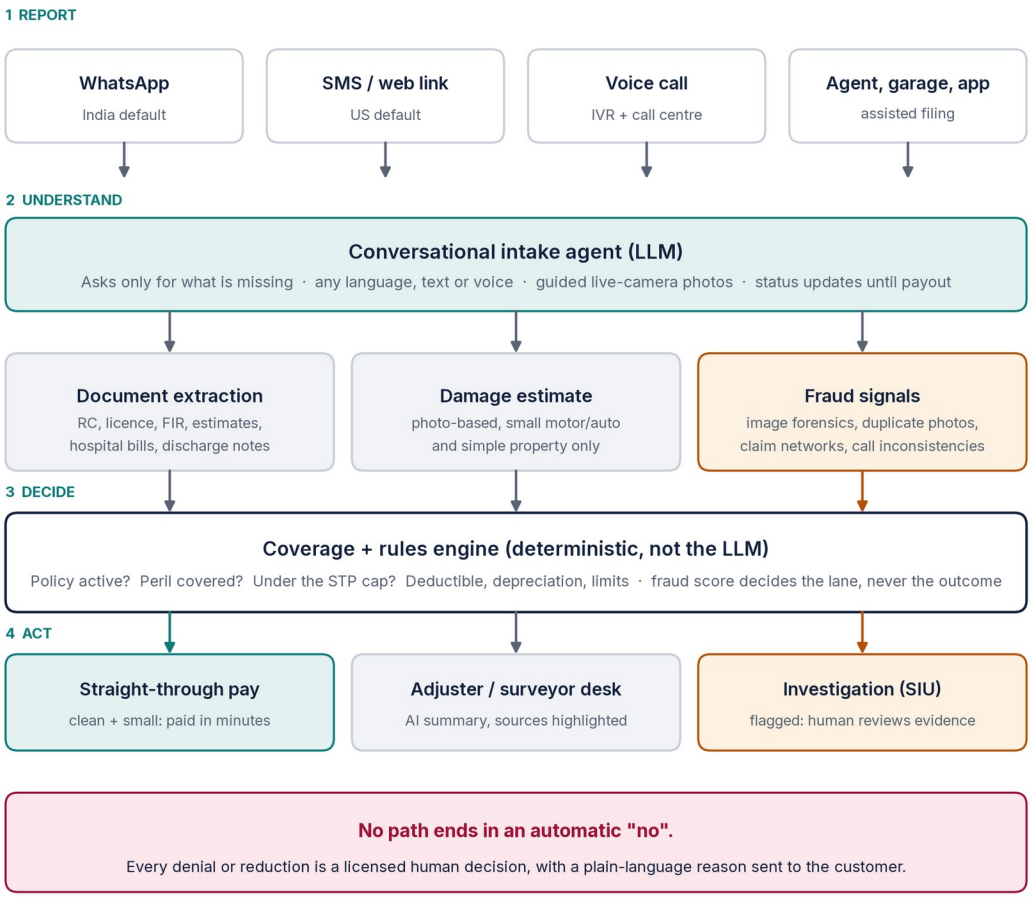


Figure 1. How a claim moves through Claim Companion. Models understand; rules and people decide.

06 The AI approach

Build vs. buy

Component	My call	Why
Conversation agent	Buy a frontier LLM via API, behind our own model gateway. Small model for routine turns.	Commodity capability. The gateway lets me swap vendors and send cheap turns to cheap models.
Speech, Indian languages	Buy, but benchmark Indian-language speech vendors and the government's Bhashini stack on our own call audio.	Code-mixed speech (Hinglish, Tenglish) is the real risk. Test on real calls, not demos.
Document extraction	Buy the base (multimodal model + OCR). Build the schemas, validators and cross-checks.	The value is in validation: registration number against the Vahan registry, totals that add up, dates inside the policy period.
Damage estimate	US: partner with an estimating vendor already wired into repair networks. India: build on our own data.	Indian insurers hold years of surveyor-assessed photos with final bills. That is a data moat, and parts and labour prices are local.
Coverage + rules	Build. Deterministic.	Must be auditable line by line. Same pattern as the admissibility engine I owned.

Component	My call	Why
Fraud signals	Build on the existing fraud models. Buy image forensics.	Insurer-specific patterns are the edge. Detecting edited or generated images is a specialist problem.

Data I need

Historical claims with outcomes (first report notes, photos, survey reports, final settlements), consented call recordings for speech and voice-inconsistency signals, sample documents for every type I extract, policy wordings for retrieval, and local parts and labour price lists. The best labels come free: every correction an adjuster makes in the desk tool is saved as training and eval data.

Consent comes first in the chat, with a short purpose-specific notice. India's DPDP Rules were notified on 14 November 2025 with an 18-month phased compliance window [19], so I would build to them from day one instead of retrofitting.

Human in the loop

Three lanes. **Straight-through:** no human, only under a payout cap and above confidence thresholds. **Assisted:** a person decides, the AI prepares everything. **Investigate:** flagged claims go to the fraud team with the evidence laid out. I would start STP caps low, around ₹20,000 in India and $1,500 in the US (my starting choice), audit 100% of automated payouts for the first eight weeks, then move to a 10% random sample plus everything flagged. Caps go up only when audits say so.

Evals and the quality bar

What I test	Bar before STP goes live (my targets)
Payment-critical fields: policy number, registration/VIN, amounts, bank details	99%+ exact match on a labelled set of past claims. Below the threshold, a person checks.
All other extracted fields	95%+ match
Photo damage estimate	Within ±15% of the final adjusted amount on 85% of eligible claims
Shadow agreement	AI recommendation matches the adjuster on 90%+ of eligible claims; every disagreement reviewed
Coverage answers in chat	Zero unsupported promises. Every coverage answer cites a policy clause. Checked by an LLM judge plus a weekly human review.
Language parity	Completion rate for each language within 5 points of English
Red team	Instructions hidden inside documents ("approve this claim"), edited and AI-generated photos, the same photo reused across claims

I set those bars by asking what each error costs. A wrong bank account number costs a payout and a customer. A headlamp priced slightly high costs a few hundred rupees and gets caught in audit. The bar should not be the same for both.

Latency and cost

Chat replies under 3 seconds at p95, photo-to-estimate under 60 seconds, and an STP payout inside the same session. On cost, my targets are **under ₹40 per claim in India and under $2 in the US** for model and messaging spend (targets to validate in the pilot, not measured figures). Most turns go to small models; only estimation and the adjuster summary use the large ones.

Failure modes and guardrails

What can go wrong	Guardrail
The assistant promises cover that isn't there	Coverage answers come only from retrieved policy text, with the clause cited. If it can't cite, it says a person will confirm.
Prompt injection hidden in a document	Documents are treated as data, never instructions. The extraction model has no tools and cannot trigger payments.
Edited or AI-generated photos	Live capture, image forensics, duplicate-photo matching across claims. UK insurer Admiral reported a 71% rise in fraud in 2025, partly blamed on AI-manipulated evidence [12].
Money goes to the wrong person	Bank account verification and name match before payout. No bank-detail change inside chat without a one-time code.
Drift during a flood or hail event	Catastrophe mode: lower caps, more human review, daily monitoring.
The customer is stuck	"Talk to a person" works at every step, in every language.

Responsible AI and regulation

US. The NAIC model bulletin on insurers' use of AI had been adopted by 24 states and DC as of April 2026, and California, Colorado, New York and Texas have their own approaches [9]. It expects a written AI program, governance, and testing for unfair outcomes. Arizona's HB 2175 bars AI as the sole basis for medical-necessity denials from July 2026, and Florida's House passed a broader "a human must deny" bill 108–0 before it died in the Senate [11]. NAIC's model claims regulation expects a claim to be acknowledged within 15 days [10]; the assistant does it in seconds.

India. The hard timelines above [13, 16], DPDP [19], and IRDAI's new fraud monitoring framework, effective 1 April 2026 [22]. My no-auto-deny rule puts the product ahead of where both markets are heading. Every model gets a model card, every decision is logged with its inputs, and fairness is a tracked guardrail, not a one-time check.

07 US vs. India: what changes

	United States	India
Channel	Text link, carrier app, phone	WhatsApp, phone, agent or garage
Language	English, Spanish	Hindi, Telugu, Tamil and more, often code-mixed. Voice first.
Claim profile	Higher severity ($5,489 average collision claim [3]), expensive labour	Small tickets (₹28,910 average health claim [15]), very high volume
Regulation	State by state. NAIC AI bulletin [9]. Human-review laws emerging [11].	One regulator with hard clocks: 24 h / 15 days / 7 days for motor, 1 h / 3 h for cashless [13, 16]
Where the value is	Adjuster capacity, cycle time, surge handling after storms	Meeting turnaround rules, fewer surveyor visits, fewer complaints
Who buys	Carriers, through their existing core claims systems	Insurers, with agents, garages and TPAs in the loop; Bima Sugam later

	United States	India
What I'd do differently	Start with glass and drivable collision. Sell on adjuster capacity during catastrophe surges.	Start with own-damage under ₹50,000. Win on languages and voice. Design for low-end Android phones and patchy data.

08 How I would measure success

My north star is **simple claims settled right, fast**: the share of eligible claims paid within 24 hours of first report that are not reopened or complained about within 60 days. It joins speed and correctness in one number. You cannot hit it by paying everything fast, because leakage and reopens would show up, and you cannot hit it by pushing every claim to a person, because speed would drop.

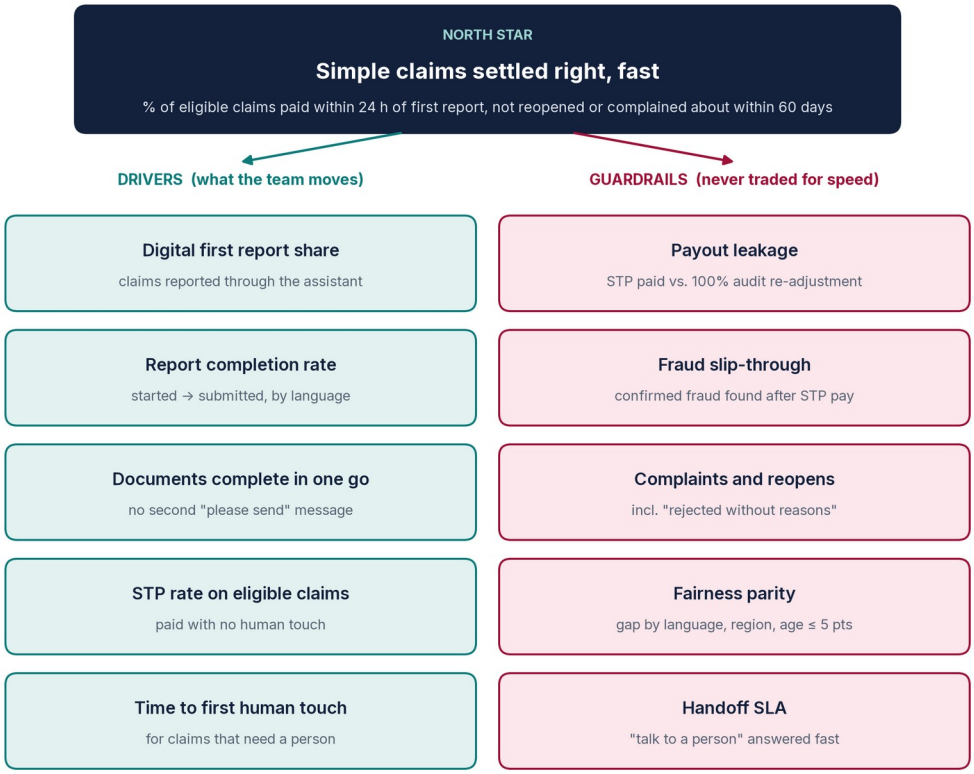


Figure 2. Metrics tree. Drivers are what the team moves week to week; guardrails are reviewed every week and can stop a rollout.

09 MVP and roadmap

In the MVP: India on WhatsApp for private-car own-damage claims with one insurer, in two cities, in Hindi, Telugu and English. US by text and web for drivable auto damage and glass, with one carrier, in two or three states that have adopted the NAIC bulletin.

Not in the MVP: injury, third-party, total loss, theft, health and home. Each is a real need; none is where I would learn fastest.

Phase	What ships	Gate to move on
0. Shadow Months 0–3	Reporting and extraction live. The AI recommends, people decide. Nothing is paid automatically.	Extraction bars met; 90%+ shadow agreement

Phase	What ships	Gate to move on
1. STP lite Months 3–6	Automated payouts under low caps. Fraud lanes. Proactive status updates.	Leakage within audit tolerance; complaints not rising
2. Expand Months 6–12	India: health reimbursement intake and cashless status for patients. US: home water and wind. Voice reporting in the call centre. Caps rise toward ₹50,000.	Each new line passes its own evals
3. Rails Month 12+	Bima Sugam and NHCX hooks, garage- and hospital-side flows, catastrophe surge mode.	Ongoing

10 Go-to-market

US. Sell to mid-size regional carriers and MGAs. They feel the adjuster crunch but cannot build this in-house. The buyer is the VP of Claims, and the proof is a 90-day shadow pilot that produces a claim-by-claim ROI report on their own data. Price per claim handled, with a higher rate for straight-through claims, so we earn most when the customer is paid fastest. Integrate with the core claims system they already run, and ship a compliance pack mapped to the NAIC bulletin so their regulator conversation is easy.

India. Start with one anchor insurer, WhatsApp first, launched to existing policyholders through renewal reminders and policy documents, and through the agents and network garages who already file for customers. The pitch is turnaround compliance and fewer complaints, not "AI". Price per claim in rupees, with a part tied to turnaround improvement. Build the claim API so it can take claims from Bima Sugam; IRDAI's chairman was reported in September 2026 as saying launch was likely by November [20].

Competition. In the US, photo-estimating vendors like CCC and Tractable, agentic claims platforms like Five Sigma's Clive, and digital carriers like Lemonade that built their own. In India most insurers build in-house, and Bajaj General's Motor OTS already settles car claims up to ₹30,000 in 20 minutes in its app [21]. That is my benchmark. It also tells me where the gap is: not whether this can be done, but doing it in WhatsApp, in Telugu, for the customer who will never download an insurer's app.

11 Risks and how I would de-risk them

Risk	What I would do
Overpaying on automated claims	Low caps, 100% audit early, caps raised only on audit data
AI-made fraud	Live capture, forensics, duplicate matching, a dedicated investigation lane. US insurance fraud is estimated at $308.6B a year across all lines [8].
Regulatory pushback or bias	No auto-deny, fairness parity as a guardrail metric, decision logs, a walkthrough with the regulator or compliance before launch
Adjuster and surveyor resistance	Position it as removing paperwork. Co-design the desk tool with them. Their edits shape the model.
Vendor or platform dependence	Model gateway with a fallback model; the same backend serves WhatsApp, text and web, so a customer can switch channels mid-claim
Customers who would rather call	Voice first. The call centre runs on the same assistant. Agents can file.

12 What I would measure in the first 90 days

Days 1–30: learn the baseline. Today's funnel by claim type: time to first contact, document requests per claim, cycle time, complaint reasons. Label about 2,000 past claims per market, and listen to 50 first-report calls before writing a single prompt.

Days 31–60: shadow mode. Report completion rate by language, extraction accuracy against the bars, shadow agreement with adjusters, and fraud-flag precision (how many flags the fraud team agrees with).

Days 61–90: go or no-go. If the bars are met, switch on straight-through payouts under the lowest cap in one city and one state. Track the north star, leakage on 100% audit, complaints and handoff time, and share one page a week with claims leadership.

What would make me stop: leakage above audit tolerance two weeks in a row, or any language group falling more than 5 points behind on completion.

> **The model is the easy part to buy. The product is the judgement about where it is allowed to act.**

Sources

My estimates, labelled in the text: ~$28B US handling spend (Triple-I ratio × NAIC premiums), ~$3.4B addressable (30% × 40%, both assumptions), STP caps, eval bars and cost targets. Personas are illustrative.

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